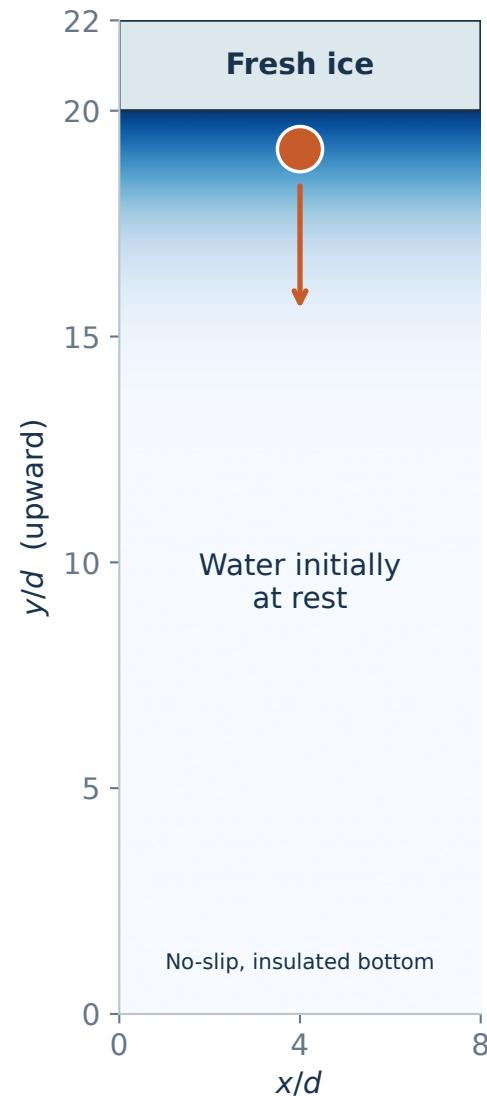


P1b | A grain settling beneath an ice shelf

Scenario schematic · initial condition · one resolved 0.2 mm quartz sphere · melting active



Periodic in x and z

A local snapshot of the meltwater layer

700 dbar · far-field CT = 0 °C · SA = 34.85 g/kg

BOX

$8d \times 22d \times 8d = 1.6 \times 4.4 \times 1.6 \text{ mm}$

INITIAL STATE

Ice interface: $y/d = 20$

Grain centre: $(4, 19.15, 4)d$; top clearance: $0.35d$

Interface water: -1.670 °C , 21.30 g/kg

WHAT WE MEASURE

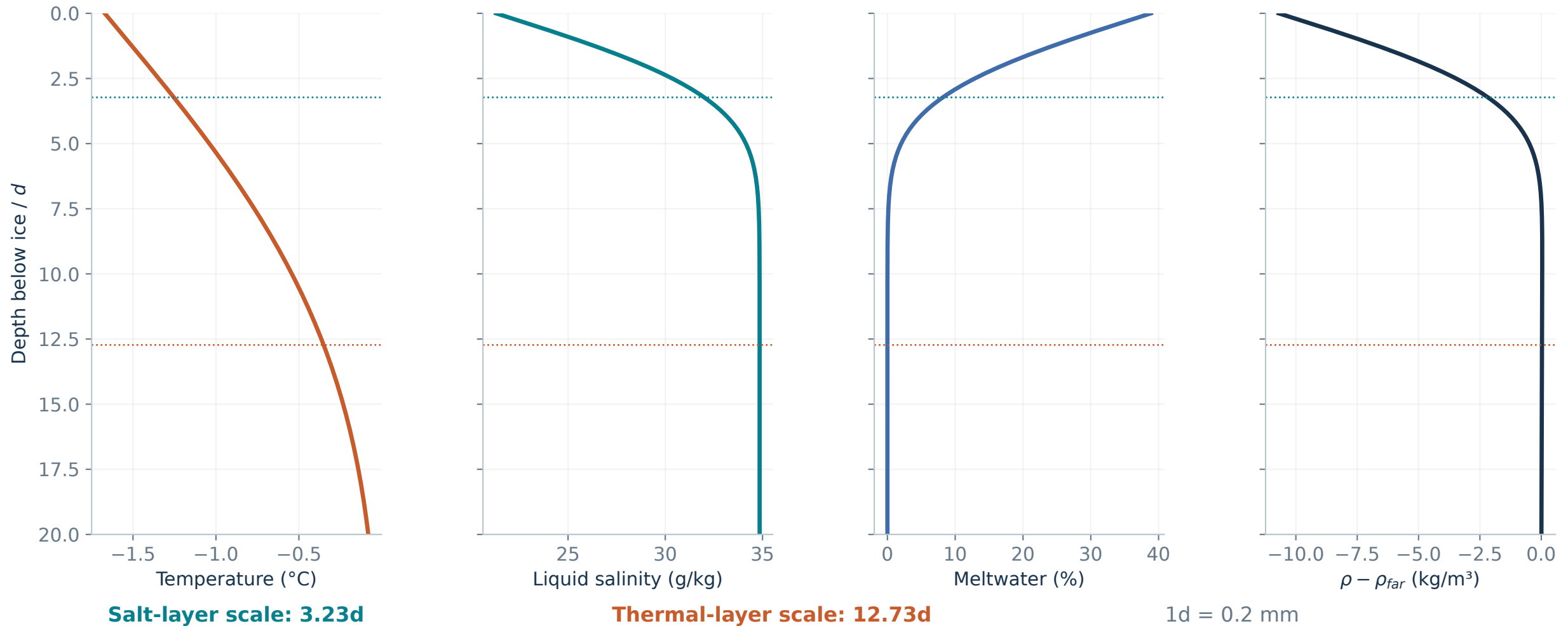
Settling speed · meltwater carried downward

Wake evolution · scalar transport · continued melting

Schematic is to scale in x-y; shading indicates prescribed meltwater. Release is the initial condition; no background current is imposed.

Initial sublayer | Cold and fresh beneath the ice

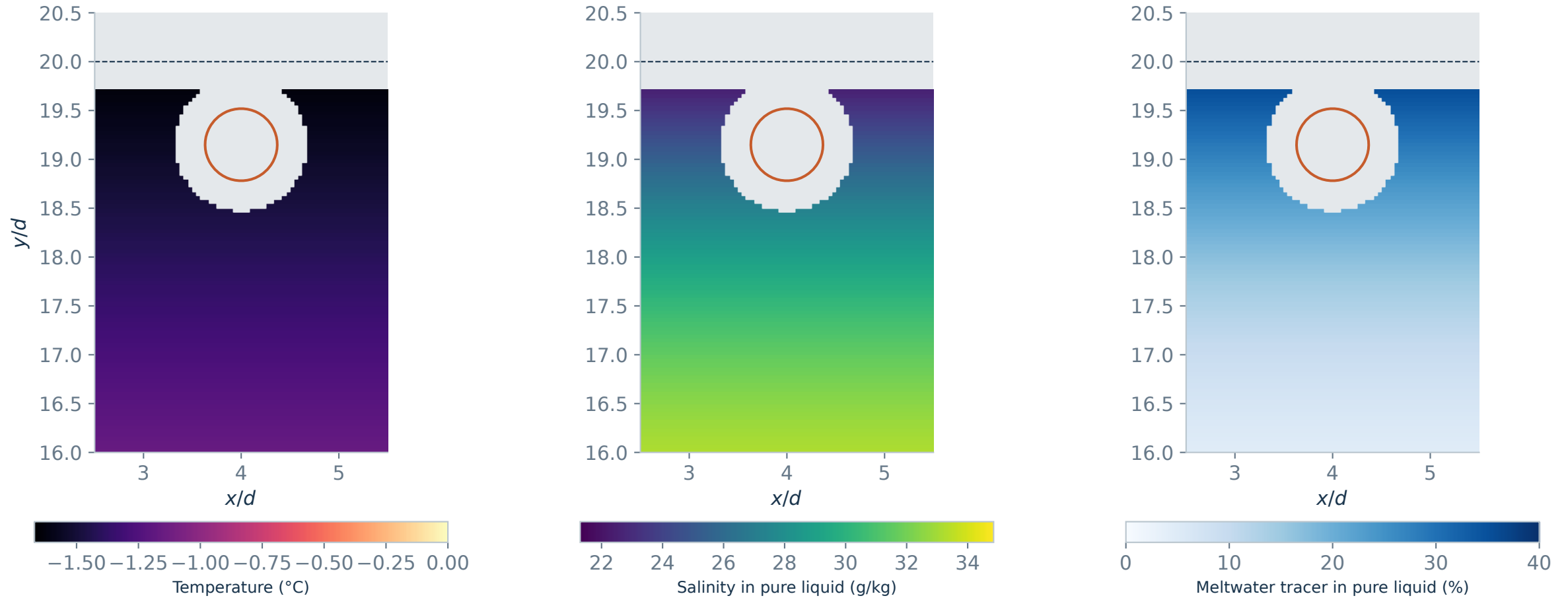
Prescribed pure-liquid profiles · depth measured downward from the initial ice interface · no imposed turbulent forcing



Analytical initialization, before diffuse-band weighting. Density uses the configured nonlinear EOS; far-field values are references, not wall forcing.

Simulation fields | Latest captured grain snapshot

Job 20977443 · default salt operator · $t = 0.00000 = 0.000$ ms · $z/d = 4.0208$

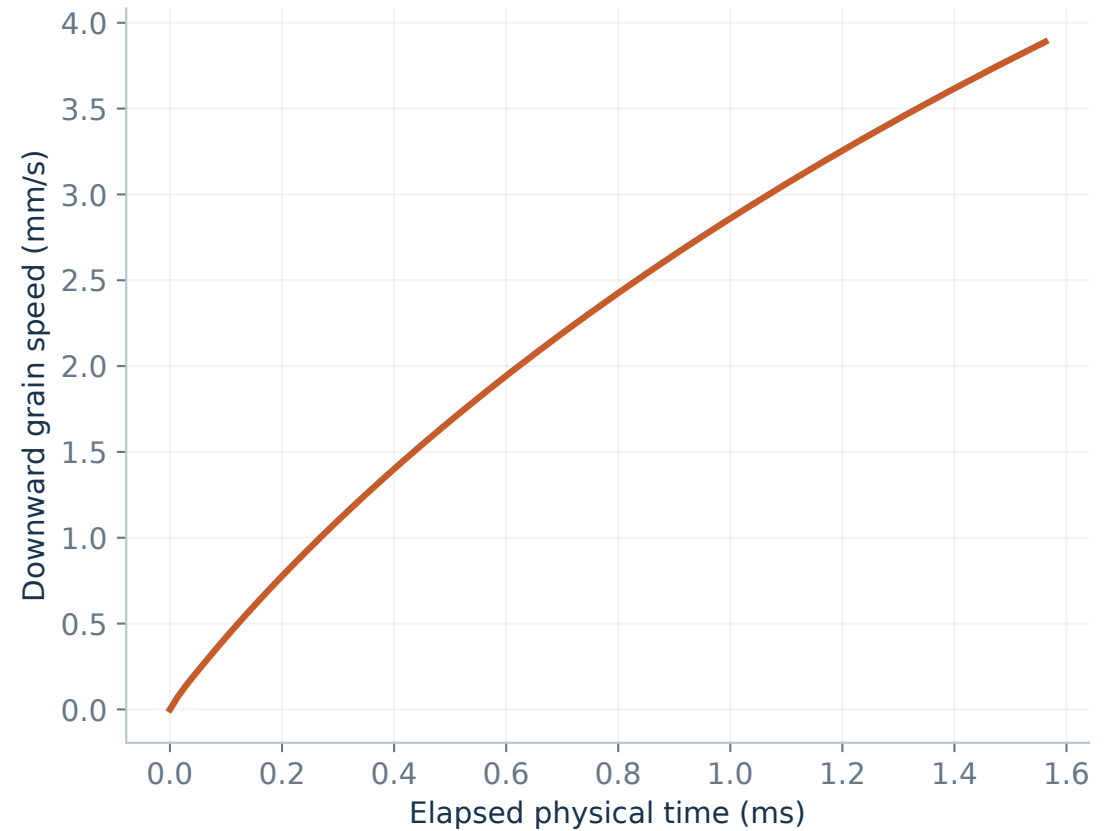
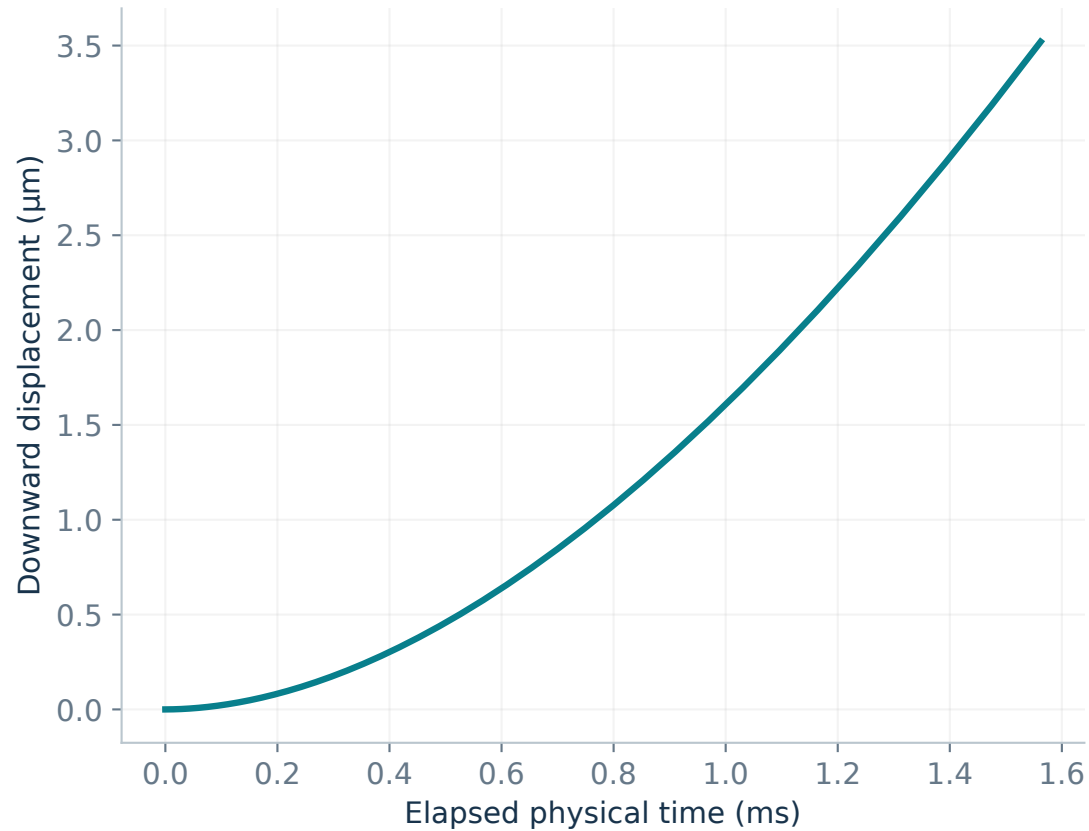


Initial output only: a developed wake is not yet available.

Actual HDF5 slices. Grey excludes ice, grain and diffuse support ($C_L < 0.999$ or $C_S > 0.001$); orange outlines $C_S = 0.5$.

Grain motion | Initial acceleration recorded so far

Job 20977443 · actual particle output · captured through $t = 0.05705$ (1.562 ms)

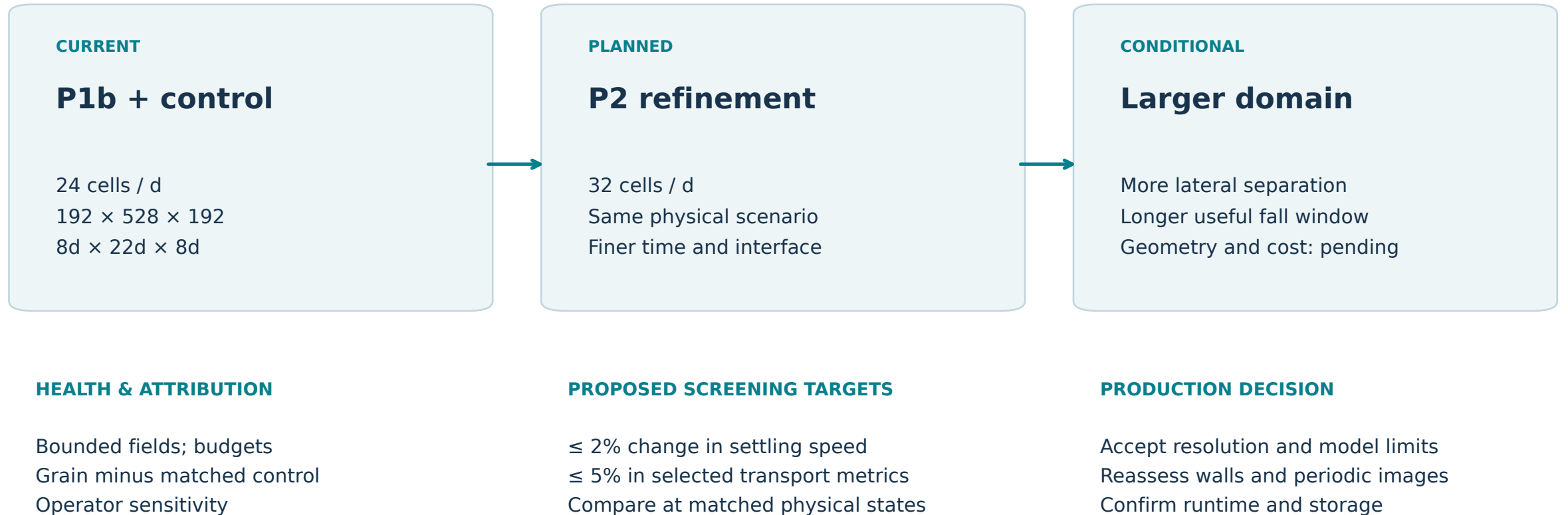


Planned observation window: 821 ms

Only the recorded interval is plotted. These curves do not establish terminal settling or precursor acceptance.

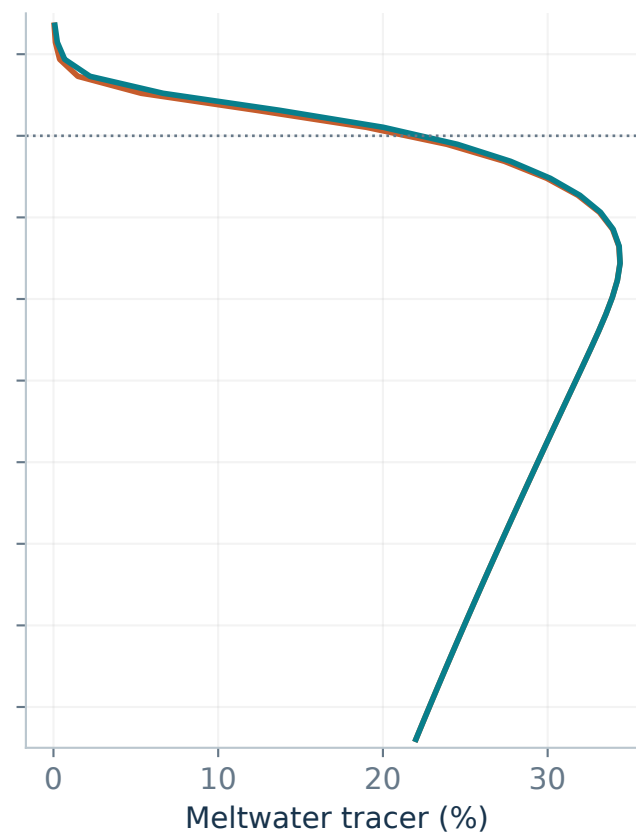
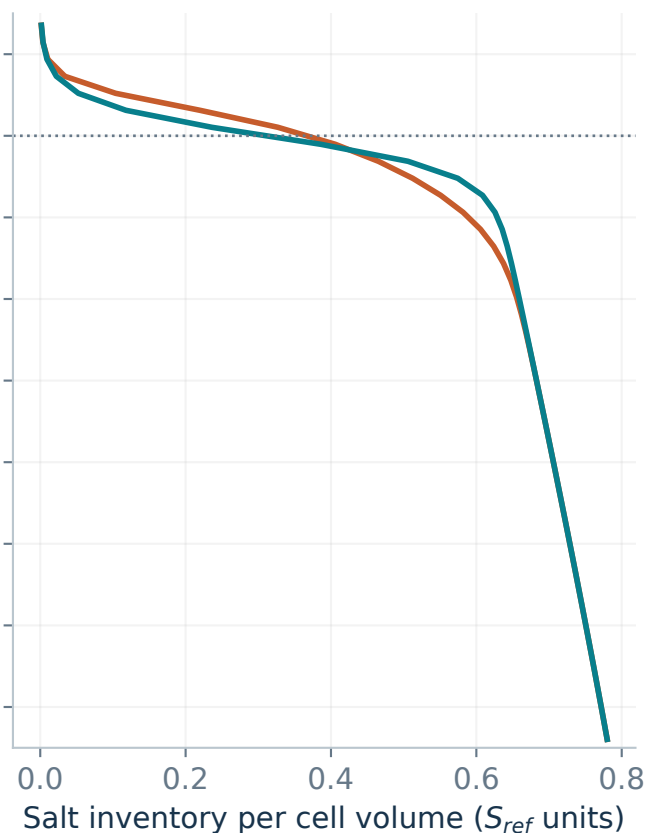
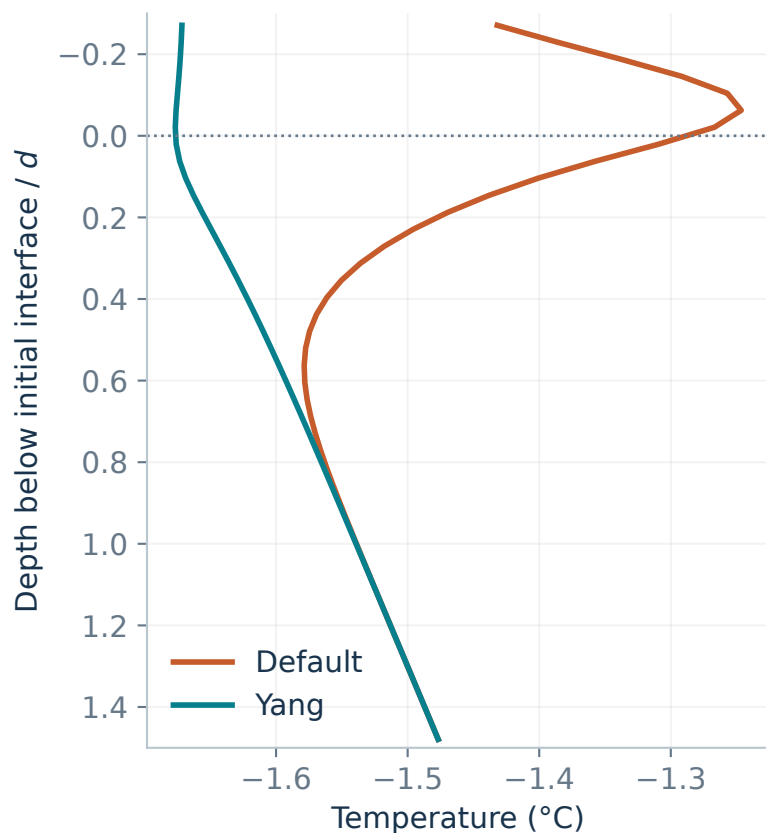
From precursor to larger-domain production

P1b establishes operability and signal size; P2 screens resolution sensitivity; production geometry remains to be selected



Ice-only controls | Early salt-operator comparison

Matched saved time $t = 0.499013 = 13.664 \text{ ms}$ · default job 20977444 vs Yang job 20973341



The comparison isolates background-layer sensitivity to the salt operator.

It does not measure sensitivity of the grain-layer interaction. This early snapshot is not an acceptance test.

Actual profiles at x, z near the periodic corner. Salt compares default s with Yang $C_L S$; it is not interface liquid salinity.